Subjective Poverty and the Limits of Relative-Income Measurement: Evidence from Greece, 2015–2024

Working paper

ABSTRACT

Greek households report difficulty making ends meet at a rate roughly 53 percentage points above the country's official at-risk-of-poverty rate, a divergence sustained over a decade and far larger than in any other member state. We ask what that gap is composed of. Using a country-year panel of the EU27 over 2015–2024 and a pre-specified testing protocol, we establish three things. First, the divergence is not an artefact of the broader official measure: at-risk-of-poverty-or-social-exclusion closes about a fifth of it, and its contribution is declining. Second, reported difficulty co-moves with concrete affordability failure within countries, though all such items are drawn from the same survey instrument as the outcome and therefore corroborate rather than validate it. Third, material resources, long-term unemployment and wage-adjusted affordability each predict reported hardship beyond income poverty, and accumulated exposure on labour, wages and housing retains cross-country predictive information after present-day conditions are controlled. We find no evidence supporting a within-country dynamic reading, and the within-country estimates are too imprecise to exclude one. A single specification choice — whether to admit a same-instrument deprivation predictor — moves Greece from the third to the twenty-fifth largest unexplained residual in the sample, and we report both specifications without selecting between them. Most of the gap remains unexplained.

Keywords: subjective poverty; material deprivation; relative income thresholds; anchored poverty; Greece; EU-SILC

1. Introduction

The European Union measures poverty in two ways that are both official and rarely compared. The at-risk-of-poverty rate (AROP) counts households below 60% of their own country's median equivalised disposable income. A separate EU-SILC item asks households directly whether they have difficulty making ends meet. The first is a position in a national income distribution; the second is a report about lived circumstances. There is no reason the two must coincide, and across most of Europe they diverge modestly and stably.

Greece is the exception. Over 2015–2024 the average distance between the two measures is 52.6 percentage points. Greece ranks first of twenty-seven member states on reported hardship while ranking seventh on AROP, and the distance between Greece and the next-ranked country on the subjective measure exceeds the range spanning most of the remaining distribution. This is not the tail of a continuum; it is a separation.

This paper asks what that separation consists of. The question admits several answers that are not mutually exclusive, and the contribution here is to test them under one pre-specified protocol rather than to advocate for any of them. The candidate accounts are: that the official measure is too narrow, and a broader one would close the gap; that the relative income threshold itself moved during the Greek collapse, so that AROP could not register a decline affecting the whole distribution; that households are reporting real material difficulty that income-based measures do not capture; that present-day economic conditions carry information beyond income poverty; that the *duration* of adverse conditions carries information beyond their present level; and that Greek respondents answer subjective questions more negatively than others.

Our answer is layered and incomplete, and we state the incompleteness as a result rather than as a caveat. The measurement accounts — a broader concept of poverty, and a threshold that fell with the economy — are real and account for a minority of the distance. The reported hardship corresponds to concrete affordability failure, though the evidence for this comes from within a single instrument. Three present-day conditions and three accumulated ones carry predictive information beyond income poverty. A reporting tendency cannot be excluded. And after all of this, most of the 52.6-point gap is not attributed.

The contribution is threefold. First, we decompose a specific and unusually large measurement divergence rather than documenting it, testing each available account under a protocol fixed before the results were seen. Second, we distinguish throughout between constructs the design excluded and constructs it could not resolve, and we compute the minimum detectable effect for each negative result so that the distinction is quantitative rather than rhetorical; this matters because the majority of our negative results fall into the second category and would be misreported as nulls under conventional practice. Third, we report the model dependence of a central result at full strength rather than selecting the specification that supports the argument, and we report two pre-specified designs that failed.

The corresponding limitation should also be stated at the outset. This is a country-level analysis of twenty-seven countries over a decade. It cannot identify causal effects, cannot speak to individual households, and has limited power against effects of moderate size.

Readers seeking a decomposition of the Greek gap into attributable shares will not find one here; what we offer is a bounded account of which candidate explanations survive testing, and an explicit statement of how much remains unexplained after they do.

Three features of the design deserve statement at the outset, because they determine how the results should be read.

The protocol was fixed before testing. Nine present-day constructs and eight accumulated ones were specified in advance, together with the conditions each had to satisfy and the order in which those conditions applied. Verdicts are the output of that procedure rather than a reading of the resulting coefficients. This matters most for the negative results: it is what allows a distinction between constructs the design could not resolve and constructs it could have detected and did not.

Inference is cross-country. The unit of analysis is the country-year. Nothing in this design identifies a causal effect, and nothing here describes an individual household. We use "predicts" throughout in its statistical sense and avoid causal language even where a causal reading would be natural.

One result is model-dependent, and we report it as such. Whether to admit a same-instrument deprivation predictor is a defensible choice either way, and the two choices place Greece at opposite ends of the residual distribution on identical rows. We present both and select neither, because selecting on the size of a residual is the practice that makes robustness checks uninformative.

The paper proceeds as follows. Section 2 describes the Greek institutional and macroeconomic setting. Section 3 reviews the relevant literature on relative thresholds, subjective measures and the Greek crisis. Section 4 sets out the data and the testing protocol. Section 5 establishes the paradox and examines the two measurement accounts. Section 6 reports what predicts reported hardship, and what does not. Section 7 discusses competing explanations that this design cannot adjudicate. Section 8 states the limitations. Section 9 concludes.

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2. Institutional and macroeconomic background

Greece entered the euro area in 2001 following a decade of macroeconomic stabilization, and experienced a decade of debt-financed, consumption-driven growth thereafter, averaging 4.1% real GDP growth in 2000–2007 (Pagoulatos, 2018). This growth was underwritten by persistent primary fiscal deficits, a near-tripling of private-sector debt as a share of GDP, and a current-account deficit that reached 15% of GDP by 2008 — financed overwhelmingly by external capital inflows rather than domestic saving or foreign direct investment (Pagoulatos, 2018). The 2008 global financial crisis triggered a "sudden stop" of these inflows; a 2009 revision of the reported budget deficit from an official 6% to an eventual 15.1% of GDP eroded market confidence sharply, and by early 2010 Greece had lost market access entirely.

What followed was, on any conventional metric, one of the deepest peacetime economic contractions on record for an advanced economy: an eight-year-equivalent depression as steep as the United States' in the 1930s, but roughly twice as long (Pagoulatos, 2018). Real GDP fell by around a quarter from its 2008 peak; unemployment peaked at 27.8% in 2013 (Eurostat series for ages 15–74, used throughout this paper; Pagoulatos reports 27.5% on a different series vintage); three consecutive financial-assistance programs were required between May 2010 and August 2018, a distinction unique among euro-area crisis countries. A brief account of these programs' design and record is given in Section 7.3, as one contextual dimension for interpreting this paper's own empirical findings rather than as background scene-setting; readers uninterested in that connection can proceed directly to Section 3.

2.1 A contraction measured in years as well as depth

The depth of the Greek contraction is widely reported; its duration is the feature that motivates the measures in Section 6.3. Unemployment did not return below 20% until 2019, eleven years after the peak of the cycle, and long-term unemployment — those out of work for a year or more — stood at 16.4% of the labour force as late as 2015 and did not fall below 10% until 2021. For a substantial cohort, joblessness was not an episode between jobs but a condition lasting the better part of a decade.

Recovery in the aggregate has been real and rapid. Unemployment fell from 25.0% in 2015 to 10.1% in 2024, and long-term unemployment from 16.4% to 5.4% over the same period. Actual individual consumption per head rose from approximately 14,800 to 21,300 purchasing power standards. On the headline labour and consumption series, Greece in 2024 is a substantially different country from Greece in 2015.

2.2 What has not returned

Two series behave differently, and they are the ones this paper's accumulated measures track. Real wages stood at roughly 77% of their 2008 level in 2015 and at 68% in 2024: they did not recover across the observed period, and by this measure Greek wages have now been below their pre-crisis level for longer than in any other member state except Hungary. Housing cost overburden — the share of the population spending more than 40% of disposable income on housing — fell from 45.5% in 2015 to 28.9% in 2024, an improvement from a level that remained several times the EU median throughout: 5.2 times it in 2015 and 4.3 times it in 2024.

This divergence between the labour aggregates and the wage and housing series is the substantive reason a paper about Greek hardship cannot rely on present-day conditions alone. A household observing an unemployment rate of 10.1% in 2024 is also observing a wage that has not recovered in sixteen years.

2.3 Composition

The population itself changed over the period. Net emigration of Greek nationals through the crisis years was substantial and has partially reversed since; the aggregate series used here cannot separate a country whose circumstances improved from a country whose composition changed, and Section 7.3 records this as an unresolved interaction rather than a controlled factor.

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3. Related literature

3.1 The moving-threshold problem

The mechanism at the center of this paper's first research question — that a relative poverty line can understate a crisis because the line itself falls with the collapsing average it is drawn from — is well established, including for Greece specifically. Andriopoulou, Kanavitsa, and Tsakloglou (2019/2020) compute an anchored (fixed-threshold) poverty rate for Greece for 2007–2016 using household-level EU-SILC microdata, finding that a 2007-anchored measure shows substantially more deterioration than the contemporaneous relative AROP series; their income-year-2013 anchored rate of 48% of the population is used in Section 5.4 as an external validation benchmark for this paper's own, coarser reconstruction. Leventi and Matsaganis (2016), in an OECD working paper using the EUROMOD microsimulation model, independently document the same arithmetic: Greece's relative poverty rate moved only modestly across 2009–2014 (13.2% to 13.8%) despite a collapse in absolute living standards over the same years, precisely because the underlying median income fell in step. The OECD's own 2018 Economic Survey of Greece reaches the same conclusion through a different lens, noting that falling wages mechanically lowered the relative poverty threshold during the crisis, making some groups appear relatively better protected even as their real living standards deteriorated (OECD, 2018). Goedemé et al. (2022) propose a new EU-wide poverty indicator, the extended headcount ratio, designed specifically to address this same measurement problem, and use Greece as their lead illustrative case.

3.2 Subjective poverty as a research object

Subjective poverty measurement is an active, distinct literature with its own methodological conventions. Želinský, Mysíková, and Garner (2022) track subjective income-poverty rates across the EU using a "minimum income question" design, a different construct from the "ability to make ends meet" measure used here and by Eurostat's own headline indicator, but one that independently flags Greece and Bulgaria as extreme outliers. A related paper (Želinský, Ng, & Mysíková, 2020) derives an income threshold from make-ends-meet responses using the Youden index — a genuinely different construction from this paper's anchored-poverty measure, and a natural extension for future work (Section 8). Guagnano, Santarelli, and Santini (2016) model subjective poverty across Europe against household and community characteristics including social capital, a reminder that economic variables are not the only plausible predictors in this literature.

3.3 Cross-country gaps beyond economic fundamentals

The central empirical puzzle of this paper — a country reporting persistently worse subjective outcomes than its measured economic conditions predict — has a structural parallel in the "happiness gap" literature comparing post-communist and Western European countries. Nikolova (2016) finds that gap is not fully explained by economic conditions, and that institutional quality (rule of law specifically) accounts for a further share of it. This is not evidence about Greece, and the outcome variable (life satisfaction) differs from subjective poverty, but it motivated this paper's own tests of household debt and welfare-transfer effectiveness as candidate residual explanations (Section 6.2), both of which returned null results. Separately, Claessens, Kyritsis, and Atkinson (2023) demonstrate that cross-country regressions commonly treat national observations as more independent than they are, understating true uncertainty when neighbouring or culturally linked countries co-move; this paper's use of country-clustered standard errors partially, but not fully, addresses that concern (Section 8).

3.4 Housing tenure, scarring, and institutional trust

Filandri, Pasqua, and Tucci (2025) link housing tenure and rent regulation to subjective poverty among young European adults (18–34) using EU-SILC data across 24 countries, providing theoretical grounding for this paper's own housing-tenure findings (Section 6.1), though their result is age-specific and not shown here to generalize to the full population. Baldini, Gallo, and Torricelli (2020) find that a past income-scarcity spell continues to depress subjective make-ends-meet assessments up to two years later across Europe, even controlling for current income — the same scarring mechanism this paper documents for Greece over a far longer horizon (Section 6.1). Gourinchas, Philippon, and Vayanos (2016, 2017) characterize the Greek crisis as one of the most severe "trifecta" crises (sudden stop, sovereign debt, and lending boom-bust together) on record, with output collapse "significantly more severe and protracted" than any comparable post-1980 episode. On institutional trust specifically, Ervasti, Kouvo, and Venetoklis (2019) show the crisis damaged Greek trust in political and impartial institutions while leaving interpersonal trust largely intact, and Economou et al. (2014) link low institutional and interpersonal trust to worse mental-health outcomes during the crisis. On migration, Labrianidis and Vogiatzis (2013) argue skilled emigration and the debt crisis were mutually reinforcing rather than the former simply a consequence of the latter, and Pratsinakis (2022) cautions against reducing the post-2010 emigration wave to a single "brain drain" narrative, showing mixed motives among emigrants in an original survey.

3.5 The case against cross-national subjective comparison

The objection to the measure used here is established and deserves stating in its strongest form. Subjective assessments are sensitive to reference groups, to adaptation over time, and to national conventions of response; on that view a cross-country comparison of subjective items measures response culture as much as circumstance, in a way that comparison of income percentiles does not. If the objection holds in full, the divergence documented in Section 5.1 is a fact about Greek respondents rather than about Greek households.

We do not assume the objection away. Section 5.6 tests whether reported difficulty tracks concrete affordability failure, and Section 7.1 tests whether the Greek pattern is concentrated in financial domains or general across them. Both narrow the objection without eliminating it, and Section 7.2 shows that part of the Greek position pre-dates the crisis, which is consistent with a longstanding component that this design cannot separate from circumstance.

3.6 Duration, scarring and hysteresis

The proposition that the length of an adverse spell matters beyond its depth is well developed in the labour literature, where long unemployment spells are associated with persistent effects on subsequent earnings and employment that are not explained by the depth of the initial shock. The mechanism most often proposed — depletion of savings, erosion of skills and networks, and the exhaustion of household coping capacity — implies that two households facing identical current conditions may differ according to how long those conditions have obtained.

This paper tests the aggregate analogue of that proposition and finds it partially supported: accumulated exposure on labour, wages and housing retains cross-country predictive information after present conditions are controlled. We emphasise in Section 6.4 that this is not the household-level scarring result the literature describes, and that the design cannot deliver one.

3.7 Anchored thresholds and cumulative exposure as measurement choices

Anchoring a poverty threshold to a base year's real value is a long-standing device for separating changes in relative position from changes in living standards, and its properties are well understood: it answers a different question from the relative measure rather than a better version of the same one. The choice of base year is consequential and is the principal degree of freedom the method offers, which is why we fix it in advance and report that we did.

Cumulative exposure measures are less standardised. Constructing one requires three decisions — the baseline against which excess is measured, whether excess accumulates continuously or resets, and whether the quantity of interest is a stock of accumulated deficit or a duration — and the literature offers no settled convention for any of them. We specify each in advance and, where a construct admits more than one defensible construction, we report the one specified and note that alternatives point the same way without counting them as corroboration. This is a weaker position than a standardised measure would allow, and we regard the absence of such a convention as a limitation of the area rather than of this paper.

3.8 The Greek case in comparative perspective

Greece's position on subjective hardship is frequently noted and rarely decomposed. The literature on the Greek crisis has concentrated, reasonably, on its fiscal and macroeconomic dimensions and on distributional consequences measured with income-based instruments. The contribution attempted here is narrower and complementary: to ask what the persistent gap between the two official measures is composed of, and to report how much of it remains unaccounted for after the available accounts are tested.

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4. Data and methods

4.1 Panel and sources

The estimation panel is country-year, covering the EU27 over 2015–2024. The outcome is the share of households reporting difficulty or great difficulty making ends meet. From 2010 onward this is Eurostat's published series, used directly. Earlier years, used for description only, are derived from the official components of the corresponding earlier release under an aggregation rule validated on 432 overlapping country-years, of which 318 agree exactly and 114 differ by a single rounding step; none differs by more than 0.1 percentage points, and the cross-country rank correlation is 1.00 in every year. The validation establishes the aggregation rule only. It does not establish comparability across earlier national survey vintages, and no inferential result in this paper uses the extended years.

Income poverty, at-risk-of-poverty-or-social-exclusion, severe material and social deprivation, arrears, capacity to meet unexpected expenses and heating adequacy come from EU-SILC. Long-term unemployment comes from the Labour Force Survey. Prices and wages come from the harmonised price index and national accounts. Actual individual consumption per head in purchasing power standards comes from the purchasing-power parity programme. Incomes are equivalised using the OECD-modified scale throughout. Nothing is imputed: where a country-year lacks a value the observation is dropped from the specification requiring it.

> The outcome is Eurostat's official subjective-hardship indicator, extended backwards before 2010 using a constructed series validated against it on 432 overlapping country-years.
>
> *Limits.* pre-2010 provenance is ours, not Eurostat's.

4.2 Constructs

Nine present-day constructs were specified before testing: material resources, labour-market exclusion, loss against a country's own past (four measures), wage-adjusted affordability, inflation exposure, housing pressure, and proximate material hardship. For each construct supported at the present-day stage, a matching accumulated measure was constructed: the total excess exposure absorbed since a fixed baseline rather than the level today. Accumulated series are cumulative sums of annual excess over a baseline of 2008 or 2010 depending on data availability, and are constructed so that no value at year t uses information from any year after t.

4.3 Testing protocol

Every construct was required to satisfy the same conditions in the same order. It must add explanatory power beyond AROP and year fixed effects; it must survive false-discovery-rate correction within its declared family; its coefficient must lie in the declared adverse direction; it must not be mechanically or instrumentally too proximate to the outcome to be tested without circularity; and it must survive a wild cluster bootstrap.

Year fixed effects absorb Europe-wide shocks — the sovereign debt crisis, the pandemic, the 2022 energy shock — that would otherwise be attributed to whichever construct moved concurrently. Conditioning on AROP is what makes each test one of incremental information: a construct that merely proxies income poverty cannot satisfy the first condition.

Standard errors clustered on twenty-seven countries are unreliable, since cluster-robust inference is asymptotic in the number of clusters. We therefore use a wild cluster bootstrap with the null imposed, 1,999 replications. The restricted variant is not optional here: an unrestricted implementation returned $p = 0.82$ for a coefficient with a t-statistic of 9.69, which is a symptom of misspecification rather than a marginal disagreement. Multiplicity is controlled by the Benjamini-Hochberg procedure within families declared before testing; applying it across families regardless would be more conservative but incoherent, since the families address different questions.

4.4 Estimating equations

For a construct *x*, country *i* and year *t*, the present-day test estimates

$$H_{it} = \alpha + \beta\, x_{it} + \gamma\, AROP_{it} + \delta_t + \varepsilon_{it}$$

where H is the share of households reporting difficulty making ends meet and δ_t are year fixed effects. The hypothesis concerns β, and the conditioning on AROP is what makes it a test of incremental information rather than of association with poverty in general.

The conditional test in Section 6.3 adds the accumulated counterpart *a* alongside the present-day measure,

$$H_{it} = \alpha + \beta_1\, a_{it} + \beta_2\, x_{it} + \gamma\, AROP_{it} + \delta_t + \varepsilon_{it}$$

and asks whether β_1 survives. The between/within decomposition in Section 6.4 replaces *a* with a country mean and a deviation from it, following Mundlak, so that the cross-sectional and longitudinal components of the same variable carry separate coefficients. The coefficient on the country mean is the between-country association; the coefficient on the deviation is the within-country one. Reporting only a pooled estimate would blend them and permit a between-country result to be read as a within-country process.

All specifications cluster on country. Where two specifications are compared, as in Section 6.5, they are estimated on the intersection of their non-missing rows, so that a change in coefficient cannot reflect a change in sample.

4.5 Power, and the meaning of a negative result

A test that fails to detect an effect is informative only if it could have detected one. For every construct that did not satisfy the conditions, the minimum detectable effect was computed by simulation rather than by formula: the observed panel structure is retained, an effect of known size is injected, and the proportion of simulated samples in which the protocol recovers it is recorded. The minimum detectable effect is the smallest injected effect recovered in at least 80% of samples.

This licenses a distinction we maintain throughout. A construct is *inconclusive* where the design could not have detected an effect of the relevant magnitude, and *unsupported* only where it could have and did not — in which case the exclusion is stated at the magnitude that holds. Collapsing these two would misrepresent the majority of our negative results. For the conditional tests in Section 6.3 the minimum detectable effect is computed pair-specifically and conditionally, since the relevant question there is whether an accumulated measure adds information *given* its present-day counterpart.

4.6 Sensitivity to single countries

Each construct is additionally refitted twenty-seven times with one country removed in turn, and the range of coefficients across those refits is recorded. This guards against a result carried by a single country — including, and especially, by Greece, since a finding driven by the case the paper is about would be circular.

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5. Results I: the paradox and the measurement accounts

5.1 The divergence

Figure 1 shows the two series for Greece. They do not converge over the observed period: reported hardship begins near 80% and remains above 60% throughout, while income poverty moves within a narrow band around 20%. The distance narrows somewhat after 2016, which we return to in Section 5.6, but never approaches agreement.

FIGURE 1 **Greece reports far more hardship than countries with similar income poverty**

DESCRIPTIVE Does official income poverty account for the hardship Greek households report?

Read with this. Descriptive comparison of two official measures. Reported hardship is answered by HOUSEHOLDS and income poverty counts PEOPLE, so the axis is labelled percent rather than either. In the second view the fitted line excludes Greece, describing the European pattern Greece is being judged against rather than one Greece helped set. The labelled point marks the median country on each measure taken SEPARATELY, so it is a reference point rather than an actual country: no member state necessarily sits there. Both views are country-level and say nothing about any individual household.

How Greece's hardship gap developed

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece: reported hardship	77.7	76.8	77.2	74.1	71.0	69.8	71.0	68.4	67.0	66.7
EU median: reported hardship	28.9	25.9	22.6	20.5	18.5	16.8	15.4	17.9	16.6	17.1
Greece: income poverty	21.4	21.2	20.2	18.5	17.9	17.7	19.6	18.8	18.9	19.6
EU median: income poverty	16.3	16.5	15.9	16.4	15.4	16.1	15.7	15.6	15.4	15.5

Where countries stood in 2024		
Series	**Income poverty (%)**	**Reported hardship (%)**
Czechia	9.5	14.2
Belgium	11.4	17.1
Denmark	11.6	12.8
Netherlands	12.1	7.3
Finland	12.6	8.9
Ireland	12.7	16.9
Slovenia	13.2	13.6
Poland	13.8	11.5
Austria	14.3	13.1
Hungary	14.3	19.8
Slovakia	14.5	28.7
Cyprus	14.6	20.8
Sweden	14.8	9.9
Germany	15.5	7.3
France	15.9	21.8
Portugal	16.6	21.8
Malta	16.9	17.3
Luxembourg	18.1	8.5
Italy	18.9	18.7
Romania	19.0	23.8
Greece	19.6	66.7
Spain	19.7	21.9
Estonia	20.2	9.9
Croatia	20.3	19.8
Lithuania	21.5	11.4
Latvia	21.6	23.3
Bulgaria	21.7	37.4

> Greek subjective hardship runs 52.6 points above relative income poverty on average, and Greece ranks first of 27 on hardship while ranking seventh on AROP.

Ranking first is not by itself informative, since some country must. The relevant question is whether Greece lies at the end of a smooth distribution, in which case it is the extreme case of an ordinary pattern, or is separated from it. The distance between Greece and the second-ranked country on reported hardship exceeds the range spanning the middle of the distribution: whatever produces the Greek position is not a larger dose of what produces variation elsewhere.

5.2 One measure, or a field of measures?

A single detached indicator invites the explanation that the indicator is faulty. That explanation is weaker if the indicator has company. We therefore take twenty-five indicators of Greek economic and social conditions — wages, hours worked, prices, unemployment, migration, household debt, saving, and expectations — and for each one record whether the country falls in the worst quintile of the Union in a given year. The outcome and every model covariate are excluded from the count, so the measure cannot restate the quantity the paper sets out to explain. Years reporting fewer than ten indicators are dropped.

FIGURE 2 **On the same 16 indicators, Greece went from the EU's worst fifth on 4 in 2008 to 11 in 2024**

DESCRIPTIVE Did Greek disadvantage deepen on a few measures, or spread across many?

Read with this. Descriptive summary of the condition, not a predictor of it. DESCRIPTIVE ONLY. Breadth was tested as a predictor of reported hardship in P3a and does not survive: on its own it is not significant (p = 0.12), and once the other accumulated measures enter the model its coefficient reverses sign. It summarises the condition rather than explaining it. THE BASKET IS FIXED: these are the 16 indicators with a valid EU position in both 2008 and 2024, chosen without reference to whether they improved or deteriorated, so the same things are counted at both ends and in every year between. Unemployment, youth unemployment and the employment rate are absent because comparable EU coverage for them begins in 2009; hours worked, pay per hour, the work-effort squeeze and real wages are all present, so this is not a basket without labour-market information. The outcome and every model covariate are excluded. THE EU-COUNTRY MEDIAN is the median of the 27 member states' own shares on this basket -- a median across countries, not Eurostat's population-weighted EU aggregate -- named accordingly rather than simply 'EU'. Croatia never reaches full 16-indicator coverage in any basket year and does not appear as a line for that reason. The 'Which measures' view's axis is POSITION, not value: the indicators have no common unit, so 0 is the best place in the Union to be on that indicator and 100 the worst, whichever direction 'worse' runs in.

Of 16 indicators, **4 were already in the EU's worst fifth in 2008, 7 entered it by 2024, and none left**.

Which measures

Series	2008 position	2024 position	Transition
Hours worked against hourly pay	59.3	100.0	entered worst fifth
Prices measured against wages	48.1	100.0	entered worst fifth
Pay per hour worked	55.6	100.0	entered worst fifth
Household income after inflation	51.9	100.0	entered worst fifth
Wages after inflation	51.9	100.0	entered worst fifth
The poverty line after inflation	51.9	100.0	entered worst fifth
Keeping the home warm	73.1	98.1	entered worst fifth
Hours worked each week	94.4	100.0	already worst fifth
What households expect of the year ahead	100.0	100.0	already worst fifth
Citizens leaving the country	86.4	96.2	already worst fifth
Income inequality	80.8	81.5	already worst fifth
Food prices	14.8	77.8	outside both years
Economic output per person	55.6	77.8	outside both years
Prices overall	53.7	70.4	outside both years
Housing and energy prices	63.0	59.3	outside both years
What households actually spend	48.1	48.1	outside both years

How many measures

Series	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Greece	25.0	18.8	43.8	50.0	50.0	50.0	43.8	50.0	50.0	62.5	56.2	56.2	56.2	56.2	56.2	62.5	68.8
EU-country median	6.2	12.5	9.4	12.5	18.8	18.8	15.6	12.5	18.8	18.8	12.5	12.5	18.8	12.5	12.5	18.8	12.5

Greece was in the worst quintile on roughly a quarter of these indicators before the crisis and is in the worst quintile on approximately two thirds of them now. Sixteen of the twenty-five now place the country at or near the bottom of the distribution, and they are not restatements of a single quantity: hourly compensation, hours worked, the real value of the poverty threshold itself, the household saving rate, expectations for the coming year, and net migration of nationals all sit together.

This measure is descriptive and is treated as such throughout. It was entered as a candidate predictor of reported hardship in the exploratory Family D analysis and did not survive: alone it is not significant ($b = 5.94$, $p = 0.12$), and when the other accumulated measures enter the same specification its coefficient reverses sign ($b = -2.17$). A quantity whose sign depends on the rest of the model cannot support an explanatory reading. It is not among the six constructs of Section 6.2, which were pre-registered; this test was exploratory throughout and could not have strengthened the evidentiary tier whatever it returned. Its role here is to establish that the divergence in Section 5.1 sits inside a broad deterioration rather than standing alone, which is a claim about the setting and not about mechanism. The second panel plots each indicator by *position* in the European distribution rather than by value, because the twenty-five share no common unit; 0 is the most favourable position in the Union on that indicator and 100 the least, in whichever direction the indicator runs.

5.3 Does the broader official measure close it?

The European Union already treats income poverty alone as too narrow. Its headline social indicator, at-risk-of-poverty-or-social-exclusion (AROPE), counts a household as affected if it falls below the income line, is severely materially deprived, or lives in a household with very low work intensity. If the divergence is simply a matter of AROP being too narrow, AROPE should largely dissolve it.

> Switching to AROPE closes 9.8 of those 52.6 points (19%), leaving 42.8 unexplained, and its contribution shrinks from 11.0 points in 2015 to 7.3 in 2024.

It does not. AROPE recovers under a quarter of the distance, and the direction of travel is against the measurement account: its contribution falls from 11.0 points in 2015 to 7.3 in 2024. As a resolution of this puzzle the broader measure is weakening rather than strengthening.

5.4 What the aggregate conceals

AROPE is a union of three conditions rather than a sum, so the headline rate cannot be decomposed into its components without double-counting the overlap. Examining the components separately is nonetheless informative about why the aggregate moves less than the experience it summarises. Over the observed period the components do not move together, and neither do the age groups within them. A headline rate combining a falling component with a rising one can remain stable while the composition beneath it changes substantially.

One component behaves differently by construction. Very low work intensity is defined over working-age adults, so households composed entirely of people above working age are excluded from its denominator; any comparison of AROPE across age groups is therefore partly a comparison of which components can apply. We report the components separately for this reason rather than presenting an age-disaggregated aggregate that would obscure it.

FIGURE 3 **Where AROPE sits against the headline measures, what it's made of, and which groups moved**

DESCRIPTIVE

Where does AROPE sit against income poverty and reported hardship, what is AROPE made of, and did every age and sex group move the same way?

Read with this. These are changes in group-level rates, not evidence about the same individuals over time. The 2024-2025 national increase was driven primarily by within-group changes, especially deterioration among people aged 65+, rather than population ageing. AROPE components are a UNION and may not be summed. Very low work intensity is part of AROPE, but the project does not hold a comparable national-total series for the Components view; its available age coverage uses a different population base, and aggregate component rates cannot reconstruct the AROPE union in any case. In all four views colour marks the measure or group and dash marks the country, so each Greek line is paired with the EU median for the same thing in the same colour. The per-component views with every other member state drawn behind are in the statistical appendix.

Headline measures

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Income poverty: Greece	21.4	21.2	20.2	18.5	17.9	17.7	19.6	18.8	18.9	19.6
Income poverty: EU median	16.3	16.5	15.9	16.4	15.4	16.1	15.7	15.6	15.4	15.5
AROPE: Greece	32.4	32.6	32.2	30.3	29.0	27.4	28.3	26.3	26.1	26.9
AROPE: EU median	22.5	22.2	21.4	20.1	20.1	19.9	19.6	19.6	19.8	19.6
Reported hardship: Greece	77.7	76.8	77.2	74.1	71.0	69.8	71.0	68.4	67.0	66.7
Reported hardship: EU median	28.9	25.9	22.6	20.5	18.5	16.8	15.4	17.9	16.6	17.1

Components

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Income poverty: Greece	21.4	21.2	20.2	18.5	17.9	17.7	19.6	18.8	18.9	19.6
Income poverty: EU median	16.3	16.5	15.9	16.4	15.4	16.1	15.7	15.6	15.4	15.5
Material deprivation: Greece	17.6	18.4	18.3	16.1	15.8	14.9	13.9	13.9	13.5	14.0
Material deprivation: EU median	7.8	6.7	6.3	5.4	5.0	4.5	4.8	4.9	4.9	4.3
AROPE: Greece	32.4	32.6	32.2	30.3	29.0	27.4	28.3	26.3	26.1	26.9
AROPE: EU median	22.5	22.2	21.4	20.1	20.1	19.9	19.6	19.6	19.8	19.6

By age

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
All ages	32.4	32.6	32.2	30.3	29.0	27.4	28.3	26.3	26.1	26.9	27.5
Under 18	37.7	37.2	36.5	34.1	31.2	30.8	32.0	28.1	28.1	27.9	29.6
18-24	43.6	44.6	42.1	39.0	37.5	35.0	37.5	33.4	33.1	34.5	33.2
25-49	35.0	36.1	35.2	31.9	30.1	28.3	29.3	26.9	24.9	25.4	25.1
50-64	34.8	35.3	34.9	33.4	31.9	29.2	30.5	27.6	26.3	28.2	27.5
65 and over	17.5	16.6	18.4	19.2	20.5	19.4	19.3	21.0	23.9	24.9	27.7
EU: all ages	24.0	23.7	22.4	21.7	21.1	21.5	21.7	21.6	21.3	21.0	20.9
EU: under 18	27.4	27.1	25.1	23.9	22.8	24.1	24.5	24.8	24.9	24.2	24.3
EU: 18-24	29.7	29.8	28.1	27.2	26.6	27.8	27.3	26.5	26.0	26.3	26.3
EU: 25-49	23.3	22.8	21.3	20.3	19.3	19.6	20.2	20.0	19.6	19.2	19.1
EU: 50-64	25.6	24.4	23.3	22.4	22.1	21.5	21.9	21.0	20.8	20.7	20.8
EU: 65 and over	18.0	18.5	18.5	19.1	19.3	20.0	19.5	20.1	19.7	19.2	18.8

By sex

Series	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
All	32.4	32.6	32.2	30.3	29.0	27.4	28.3	26.3	26.1	26.9	27.5
Women	33.1	33.4	32.9	31.3	30.1	28.6	29.2	27.4	27.3	27.9	29.1
Men	31.6	31.7	31.6	29.2	28.0	26.1	27.3	25.2	24.8	25.9	25.9
EU: all	24.0	23.7	22.4	21.7	21.1	21.5	21.7	21.6	21.3	21.0	20.9
EU: women	24.9	24.7	23.4	22.7	22.1	22.5	22.6	22.7	22.3	21.9	21.9
EU: men	23.1	22.5	21.4	20.6	20.0	20.4	20.7	20.4	20.3	20.0	19.8

Two features of the disaggregation bear on the argument. Greece is at or near the top of the European distribution on income poverty, severe material deprivation and the combined measure alike, so the position is not an artefact of one component. And the burden is not evenly distributed: the age groups diverge, and women sit above men throughout, with the difference widening after 2022.

5.5 The moving threshold

The second measurement account concerns the ruler rather than the concept. AROP counts households below 60% of the *current* national median. This is a deliberate design choice and a defensible one, but it has a consequence that becomes severe in a large enough contraction: when national income falls, the threshold falls with it, and a household whose real income dropped sharply can remain above a line that dropped as sharply.

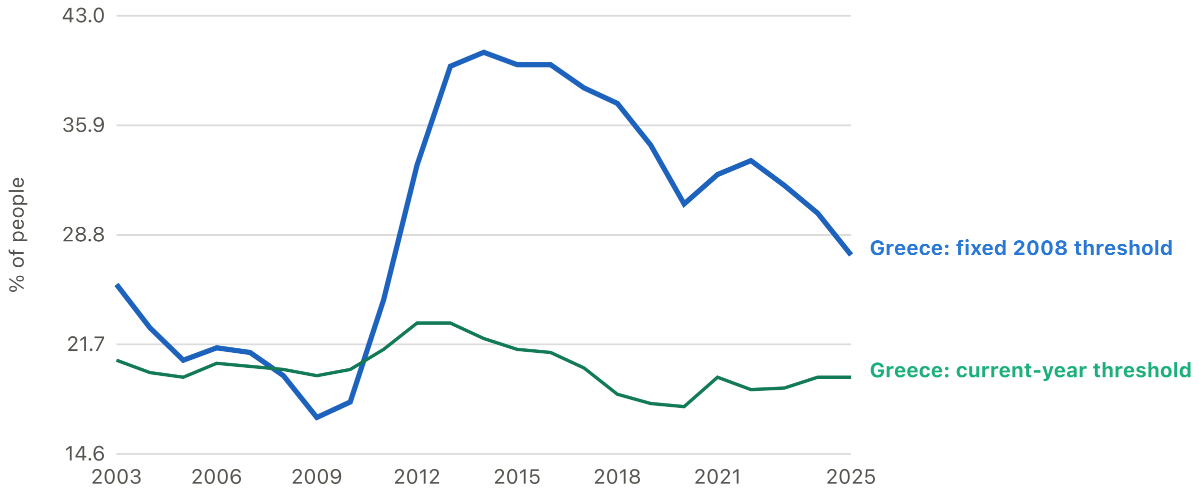
Greek median equivalised income fell by roughly a third over the crisis. The poverty line, being a fixed fraction of it, fell in step. Figure 4 compares the relative threshold with one anchored to its 2008 real value.

FIGURE 4 **Who counts as poor barely moved; what the poverty line buys fell by a fifth**

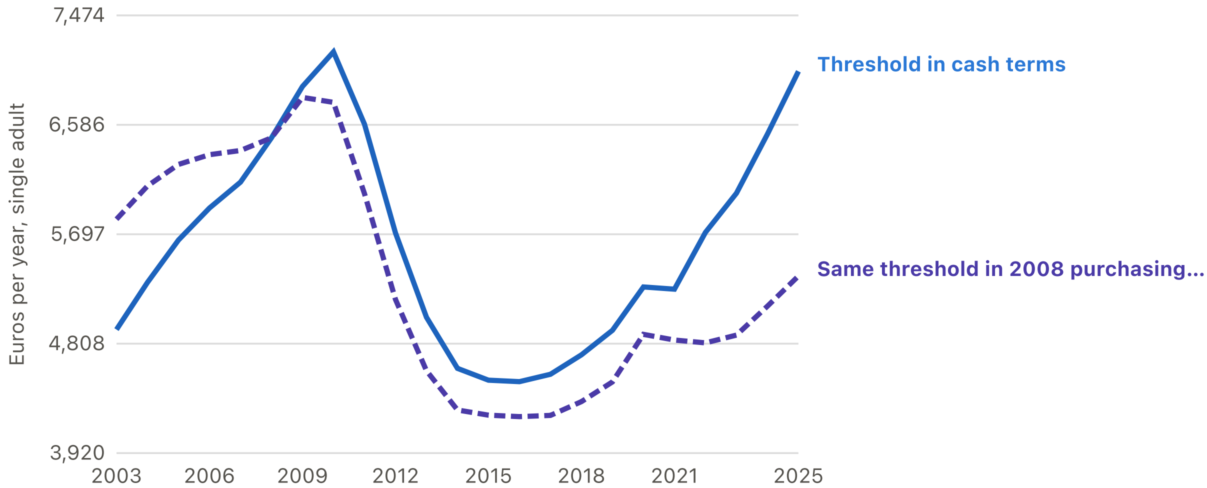
DESCRIPTIVE

What happened to the line Greek poverty is measured against, and what does a fixed line show instead?

a. Who falls below a fixed line



b. What the line itself is worth



Read with this. The anchored series is an approximation built in this project and is labelled as such wherever it appears. The first view counts PEOPLE, the second counts EUROS. In the second, both lines are the same official threshold: one as published in each year's own money, the other converted into what it could buy in 2008. The cash line recovers and the purchasing-power line does not, and that difference is why the first view's two counts diverge. Both views are Greece only and support no cross-country statement.

Who falls below a fixed line

Series	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	202
Greece: fixed 2008 threshold	25.6	22.8	20.7	21.5	21.2	19.7	17.0	18.0	24.6	33.3	39.7	40.6	39.8	39.8	38.3	37.3	34.6	30.8	32.7	33.6	32.0	30.
Greece: current-year threshold	20.7	19.9	19.6	20.5	20.3	20.1	19.7	20.1	21.4	23.1	23.1	22.1	21.4	21.2	20.2	18.5	17.9	17.7	19.6	18.8	18.9	19.

What the line itself is worth

Series	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	20
Threshold in cash terms	4923	5306	5650	5910	6120	6480	6897	7178	6591	5708	5023	4608	4512	4500	4560	4718	4917	5269	5251	5712	6030	65
Same threshold in 2008 purchasing power	5819	6089	6264	6343	6377	6480	6808	6768	6027	5168	4589	4270	4228	4216	4226	4338	4498	4884	4838	4815	4878	51

The anchor year was fixed before the comparison was run and was not selected by inspecting which choice produced the largest divergence. We note two limitations. The anchored series in this paper is constructed for Greece alone and therefore supports no cross-country statement: it shows that the Greek threshold fell relative to Greece's own pre-crisis standard, not that it fell further than any other country's. And anchored and relative poverty answer different questions, neither of which is the correct one in general. Where a whole distribution declines together, they diverge sharply, and that divergence is the substantive point rather than a defect.

Our reconstruction is coarser than the household-level anchored series computed by Andriopoulou, Kanavitsa and Tsakloglou, and we use their income-year-2013 figure as an external benchmark rather than as a replication target.

5.6 Convergence, and what it does not show

The narrowing after 2016 invites a recovery reading, and that reading does not follow. A gap between a country and the EU average can close because the country improved, because the average deteriorated, or because both moved in the same direction at different speeds. Decomposing the closure into those contributions would require attributing movement to each side, which we do not do. The measures also did not converge uniformly: some closed a substantial share of their 2015 distance and others closed little, so a single summary statement about Greek convergence would misrepresent the pattern.

5.7 Does the reported hardship correspond to anything?

If Greek households report hardship without corresponding material difficulty, the object of study is response behaviour rather than poverty, and the remaining analysis measures an artefact. We therefore test whether reported difficulty co-moves with items naming concrete events: arrears, inability to meet an unexpected expense, inability to heat the home adequately, and severe material and social deprivation.

> Reported difficulty co-moves with arrears, inability to meet an unexpected expense, inadequate heating and severe deprivation at within-country correlations of 0.63 to 0.80.
>
> *Limits.* same-instrument corroboration, NOT independent validation; all items and the outcome come from EU-SILC; not uniform: arrears within Greece is 0.371.

We report the within-country correlations because between-country correlation is weak evidence here: richer countries score better on essentially every welfare indicator, so almost any pair correlates across countries without either telling us about mechanism. The within figures are computed after demeaning each series by its country mean, so a country persistently high on both contributes nothing.

The limitation is structural and we place it in the text rather than in a footnote. Every one of these items, and the outcome, is self-reported by the same household in the same interview. A household answering the whole battery downbeat would generate correlations of this size without any item independently confirming another. This is corroboration from within one instrument, not external validation, and the pattern is not uniform: arrears within Greece correlates at 0.371, well below the range headline.

A stronger version of the same test asks how much of Greece's excess these items statistically absorb.

> Those same four items statistically absorb 71% of Greece's baseline residual (+46.92 to +13.74).
>
> *Limits.* ABSORPTION, never explanation; shared instrument is not shared cause; diagnostic only; never a headline explanation.

The word absorb is doing precise work. Once concrete deprivation items enter the model, most of Greece's unexplained excess is no longer statistically distinguishable. That is a statement about shared variance, not about mechanism, and Section 6.4 shows how much rests on the decision to admit such a predictor at all.

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6. Results II: what predicts reported hardship

6.1 Present-day conditions

Three of the nine present-day constructs satisfied every condition.

> Material resources predicts hardship beyond AROP and year effects (coef -0.0013, wild-cluster bootstrap p=0.0055).
>
> *Limits.* cross-country association; no causal claim.

> Labour-market exclusion predicts hardship beyond AROP and year effects (coef +4.3402, wild-cluster bootstrap p=0.0085).
>
> *Limits.* cross-country association; no causal claim.

> Wage-adjusted affordability predicts hardship beyond AROP and year effects (coef +0.3110, wild-cluster bootstrap p=0.0005).
>
> *Limits.* cross-country association; no causal claim.

The directions are as expected and were declared before testing: greater material resources predict less reported hardship, more long-term unemployment predicts more, and worse wage-adjusted affordability predicts more. A construct whose coefficient had pointed the other way would have been recorded as contradicting its declared direction rather than reinterpreted.

Figure 5 shows all nine as standardised effects, which is what makes them comparable: the constructs are measured in percentages, purchasing power standards and index points, and raw coefficients could not share an axis.

FIGURE 5 **Three current-condition constructs survive the full testing sequence**

PRE-PLANNED CONFIRMATORY

Which current-level constructs survive every pre-registered condition, and which gate stopped the rest?

Read with this. Inconclusive is not evidence of absence. Bars are CLUSTER-ROBUST 95% confidence intervals, not bootstrap intervals: the wild cluster bootstrap determines the final support status, and two constructs whose intervals exclude zero here still fail it.

Construct	Effect (SD)	CI low	CI high	p	p FDR	Bootstrap p
Long-term unemployment	0.808	0.523	1.093	0.000	0.000	0.009
Material deprivation	0.720	0.239	1.202	0.003	—	—
Wage-adjusted affordability	0.687	0.386	0.988	0.000	0.000	0.001
Share below own GDP peak	0.620	0.343	0.897	0.000	0.000	0.401
Material resources	0.590	0.330	0.851	0.000	0.000	0.005
Housing-cost overburden	0.545	0.141	0.949	0.008	0.015	0.553
Real household income	0.315	-0.279	0.908	0.299	0.413	—
Inflation	-0.274	-0.817	0.268	0.321	0.413	—
Real wages	0.174	-0.501	0.850	0.613	0.653	—
Real poverty threshold	0.138	-0.464	0.740	0.653	0.653	—

Table 1. Present-day constructs: coefficient, bootstrap *p*, verdict, and the condition that stopped each construct that did not clear. Coefficients are on each construct's own scale and are not comparable across rows.

CONSTRUCT	COEF.	BOOTSTRAP *P*	VERDICT	STOPPED AT
Proximate material hardship	+1.4524	—	Not testable	proximity to outcome
Loss against own past — Real wages	-0.0924	—	Inconclusive	insufficient power
Loss against own past — Real household income	-0.2168	—	Inconclusive	insufficient power
Loss against own past — Real poverty threshold	-0.0556	—	Inconclusive	insufficient power
Loss against own past — Share below own GDP peak	+1.8675	0.4005	Inconclusive	wild cluster bootstrap
Inflation exposure	-0.9925	—	Inconclusive	insufficient power
Housing pressure	+1.1385	0.5530	Inconclusive	wild cluster bootstrap
Material resources	-0.0013	0.0055	Supported	—
Labour-market exclusion	+4.3402	0.0085	Supported	—
Wage-adjusted affordability	+0.3110	0.0005	Supported	—

Leave-one-out refits confirm that the three supported constructs do not depend on any single country. Long-term unemployment ranges from 3.20 to 4.83 across the twenty-seven refits against a headline of 4.34, and wage-adjusted affordability from 0.25 to 0.33 against 0.31. Housing pressure, which does not clear the conditions, ranges from −0.10 to 1.27 and therefore changes sign when one country is dropped.

DESCRIPTIVE CORROBORATION

INDEPENDENT DESCRIPTIVE CORROBORATION (GREECE IN FIGURES)

An independently published analysis (Greece in Figures, drawing on 2025/2026-vintage Eurostat and ELSTAT releases) reaches a similar descriptive picture by a different route and with no formal test: Greece is not the EU's poorest country on actual consumption, but works among the longest hours in the Union for comparatively low hourly reward, facing high relative prices in several everyday categories. That is the same shape as material resources and wage-adjusted affordability above.

What may be concluded. The independent picture -- Greece not last on actual consumption, long hours for comparatively low hourly reward, high relative prices in food and information/communication -- is consistent with and external corroboration for this project's own material-resources and wage-adjusted-affordability findings. Its central descriptive numbers reproduce against the primary Eurostat and ELSTAT releases they draw on.

Limitation. Treating the article's own comparisons, or its constructed AIC-per-hour ratio, as inferential evidence: it applies no bootstrap, no FDR correction, no within/between decomposition and no model comparison, and it does not test whether these factors account for reported hardship. Merging its 2025/2026-vintage figures into this project's frozen 2015-2024 panel. Repeating its car-ownership sentence, which cites a vehicle-STOCK figure as evidence of new purchases, or its 'all Greeks travelled' generalisation, which the underlying ELSTAT survey does not support. Reading its AIC rank, hours rank or price-level figure as contradicting this project's own numbers without first noting the different vintage or aggregate behind each.

Greece in Figures, "Γιατί οι Έλληνες νιώθουν τόσο φτωχοί" [Why Greeks feel so poor]. Constructs an AIC-per-hour ratio from aggregate actual individual consumption and aggregate annual hours worked; cites 2025-vintage Eurostat AIC and price-level releases and a 2026 ELSTAT resident travel bulletin. link

6.2 The six that did not, and what that does and does not mean

Six constructs did not satisfy the conditions. Describing these as ruled out would misstate most of them.

> Six of nine current-level constructs are inconclusive under available power, not unsupported; two cleared FDR and collapsed under the bootstrap (p=0.40 and 0.55).
>
> *Limits.* inconclusive is not evidence of absence.

With twenty-seven countries and roughly a decade of observations the design has limited power, and for most of the six the smallest reliably detectable effect exceeds any magnitude of substantive interest. Those constructs are inconclusive: the study is silent about them rather than negative. Where the design did have adequate power, the exclusion is real but magnitude-specific.

> Annual food and housing inflation are unsupported with adequate power at 0.70 SD; annual headline inflation is unsupported at its detectable conditional magnitude; compounded inflation since 2008 remains inconclusive.
>
> *Limits.* the exclusions are narrow and magnitude-specific; compounded inflation is INCONCLUSIVE, not ruled out.

Two constructs illustrate why the bootstrap condition was imposed. Both cleared multiplicity correction comfortably and then collapsed under the bootstrap, at $p = 0.40$ and 0.55. A protocol stopping at the conventional step would have reported both as findings.

6.3 Accumulated exposure

Two countries can present identical current conditions and have reached them by different routes. Whether that history carries information is tested by placing each accumulated measure in the same specification as its own present-day counterpart and asking whether it survives. Because the two are correlated by construction, the test is conservative: shared variance is attributed to neither.

FIGURE 6 **Accumulated history provides additional conditional cross-country information in three of eight pairs**

POST-SELECTION ROBUSTNESS

Does accumulated history add anything once today's conditions are in the same model?

Read with this. cross-country association | no demonstrated within-Greece dynamic | no causal claim | current conditions not ruled out This shows the pairs that survive; all eight, including the five that do not, are in the statistical appendix. The axis is STANDARDISED: SDs of hardship per SD of the predictor. The eight measures are in different units - percentage-point-years, years, index points, percentages - so raw coefficients could not share an axis, and the raw value is given in the table and the tooltip instead. Each coefficient is tested against ITS OWN pair-specific detectable effect, not a study-wide threshold: conditional power depends on how much independent variation survives once the counterpart is controlled. Bars are cluster-robust intervals; the bootstrap decides support.

Test	Standardised effect (SD)	Raw coefficient	p FDR	Bootstrap p	Conditional MDE (SD)
Accumulated unemployment (controlling for Long-term unemployment)	0.550	0.294	0.000	0.003	0.600
Years wages below 2008 (controlling for Real wages)	0.624	2.090	0.005	0.006	0.500
Housing deterioration since 2010 (controlling for Housing-cost overburden)	0.668	0.242	0.000	0.002	0.600

> Accumulated excess unemployment predicts hardship (bootstrap p=0.0025) and retains a cross-country association after its current-level counterpart is controlled (conditional bootstrap p=0.0025).
>
> *Limits.* cross-country association; no demonstrated within-Greece dynamic; no causal claim; current conditions not ruled out.

> Duration of real wages below their 2008 level predicts hardship (bootstrap p=0.0245) and retains a cross-country association after its current-level counterpart is controlled (conditional bootstrap p=0.0060).
>
> *Limits.* cross-country association; no demonstrated within-Greece dynamic; no causal claim; current conditions not ruled out; supported only as the CURRENT uninterrupted run against a fixed 2008 base; alternative constructions point the same way but do not meet the criteria.

> Housing-cost deterioration since 2010 predicts hardship (bootstrap p=0.0460) and retains a cross-country association after its current-level counterpart is controlled (conditional bootstrap p=0.0015).
>
> *Limits.* cross-country association; no demonstrated within-Greece dynamic; no causal claim; current conditions not ruled out; BORDERLINE: bootstrap p=0.0460, 91 of 1,999 exceedances.

The third is borderline, with 91 of 1,999 bootstrap replications exceeding the observed statistic, and is reported at that strength. The second holds only under one construction — the current uninterrupted run against a fixed 2008 base — and alternative constructions of the same idea point the same way without meeting the criteria; they are not counted as corroboration.

The pattern is not uniform, which is itself informative.

> For wage-adjusted affordability the pattern reverses: the current measure survives conditioning (bootstrap p=0.0035) while the accumulated one remains inconclusive.
>
> *Limits.* the accumulated measure is inconclusive, NOT unsupported.

For wage-adjusted affordability the relationship reverses: the present-day measure survives conditioning while the accumulated one does not resolve. Were accumulation a general property of this outcome, that reversal should not appear. It suggests the accumulated results are specific to labour markets, wages and housing rather than reflecting a general history effect — and it is the clearest reason to resist a summary in which history supersedes present conditions.

One construct could not be tested at all, and we report it rather than omitting it.

> Accumulated material resources (C1) could not be tested at all: the source series begins in 2015 and no 2008 baseline exists. The baseline was not moved to make it testable.
>
> *Limits.* infeasible, not null.

> The accumulated wage-shortfall coefficient cleared every conditional gate but its prior stage did not support it, so the pre-registered ceiling caps it. It is reported and is not a finding.
>
> *Limits.* capped by the E7 ceiling: E7 may only qualify or withdraw.

6.4 The limit of the accumulated results

The results in Section 6.3 are statements about differences between countries. Converting them into statements about change within Greece over time is not supported, and Figure 7 shows why.

FIGURE 7 **The supported historical associations are predominantly between countries; the within-country estimates provide no supporting dynamic evidence**

PRE-REGISTERED CONDITIONAL ROBUSTNESS Is the effect between countries or within them?

Read with this. This is THE central limitation: no within-country estimate is significant in the adverse direction and no first-difference test supports one. The first view is STANDARDISED, and each component by its own spread: between-country and within-country variation differ in size by factors of 0.8 to 5.7 here, so one shared SD would distort the comparison it is meant to fix. The second view shows first differences in each measure's own units, which is why those bars carry no shared scale and are read one row at a time. This does NOT establish that no within-country relationship exists: the within estimates are too imprecise to establish or rule out such a relationship. They are recorded as inconclusive, not absent.

Between vs within

Accumulated measure	Between (SD)	Within (SD)	Between p	Within p	First difference (raw)	First difference p
Accumulated unemployment	0.5924	0.0296	0.0000	0.6867	0.0461	0.7324
Cumulative wage shortfall	0.6584	-0.0544	0.0130	0.3147	-0.0596	0.0284
Years wages below 2008	0.7271	0.0700	0.0025	0.0823	0.0721	0.4952
Cumulative GDP shortfall	0.3890	0.0674	0.0289	0.0600	0.0273	0.1442
Cumulative threshold shortfall	0.8271	0.0123	0.0000	0.5599	-0.0148	0.3917
Accumulated affordability pressure	-0.3953	-0.0764	0.3639	0.2825	-0.0298	0.0000
Compounded inflation since 2008	-0.1718	-0.1228	0.4947	0.3738	-0.0314	0.5766
Housing deterioration since 2010	0.9282	-0.0658	0.0000	0.2343	-0.0090	0.8679

	Year-on-year changes		
Accumulated measure	**First difference**	**p**	**n**
Accumulated unemployment	0.0461	0.7324	243.0000
Cumulative wage shortfall	-0.0596	0.0284	243.0000
Years wages below 2008	0.0721	0.4952	243.0000
Cumulative GDP shortfall	0.0273	0.1442	243.0000
Cumulative threshold shortfall	-0.0148	0.3917	234.0000
Accumulated affordability pressure	-0.0298	0.0000	243.0000
Compounded inflation since 2008	-0.0314	0.5766	243.0000
Housing deterioration since 2010	-0.0090	0.8679	242.0000

> No accumulated measure permits dynamic wording. Across P5, E4 and E7 no within-country estimate is significant in the adverse direction and no first-difference test supports one.
>
> *Limits.* three RELATED checks on one panel, not independent replications; the dynamic tests were not multiplicity-corrected.

We are explicit about what this does not establish. The within-country estimates are imprecise, and imprecision is not evidence of absence: these tests discard all cross-sectional variation and have considerably less power than the between-country ones. The correct statement is that this design does not support dynamic wording, not that a within-country relationship has been shown to be absent.

6.5 Model dependence

Whether to admit a same-instrument deprivation predictor is a genuine methodological choice. In favour: severe material deprivation is a substantively meaningful, officially published measure, and excluding variables because they are well measured is not a principle anyone holds. Against: when predictor and outcome come from the same respondents answering adjacent questions, part of any relationship reflects shared measurement rather than shared substance. We ran both.

FIGURE 8 **Removing the same-instrument deprivation measure moves Greece from under-predicted to over-predicted**

POST-SELECTION ROBUSTNESS Does the answer depend on which model is chosen?

Read with this. NEITHER specification is definitive. They may not be merged or averaged, and selection may not be made on residual size.

Country	Reference specification	Deprivation-free companion	Change
Luxembourg	8.80	15.48	6.68
Cyprus	7.05	10.75	3.70
Greece	6.93	-9.39	-16.32
Latvia	6.32	4.65	-1.66
Hungary	4.86	7.42	2.56
Bulgaria	4.06	14.65	10.60
Czechia	3.53	-3.63	-7.16
Slovakia	3.28	-0.68	-3.96
Belgium	3.13	5.26	2.13
Ireland	3.05	6.14	3.09
Croatia	2.62	-0.38	-2.99
France	1.85	3.88	2.02
Austria	1.36	0.95	-0.41
Portugal	1.30	0.62	-0.67
Malta	1.28	0.36	-0.91
Poland	0.56	-4.32	-4.88
Denmark	0.16	-3.65	-3.81
Sweden	-1.00	-4.55	-3.55
Slovenia	-2.66	-6.33	-3.67
Netherlands	-3.47	-4.87	-1.39
Lithuania	-4.17	-1.81	2.36
Germany	-4.31	-6.00	-1.69
Estonia	-4.49	-10.12	-5.63
Finland	-5.07	-6.80	-1.72
Italy	-6.35	-5.18	1.17
Romania	-10.32	9.76	20.08
Spain	-10.63	-10.06	0.57

Greece's residual is +6.93 (rank 3/27) in the reference specification and -9.39 (rank 25/27) when the same-instrument deprivation predictor is removed, on identical rows. Neither specification is definitive.

Limits. NEITHER specification is definitive; the two may not be merged or averaged; selection between them may not be made on residual size; conclusions about absorption depend materially on whether same-instrument deprivation is admitted.

Greece moves from the third to the twenty-fifth largest unexplained residual — from a marked positive anomaly to a marked negative one — on identical rows, with one predictor added or removed. This is a reversal rather than a sensitivity, and we neither average the two nor select between them. Selecting a specification on the size of its residual is precisely the practice that renders robustness checks uninformative. What can be said is that conclusions about how much of Greece's anomaly is absorbed depend materially on a choice the data cannot adjudicate.

6.6 Two pre-specified designs that failed

Two further analyses were specified in advance and did not work. We report them because a paper presenting only its successful analyses misrepresents the evidence base.

> The synthetic-control comparative design failed four of six pre-registered gates and is not a usable comparison. Its divergence figure is withheld from every document in this project.
>
> *Limits.* donor weights collapse to Hungary 0.55 and Bulgaria 0.45; the divergence figure is NON-REPORTABLE.

The synthetic-control design was intended to carry substantial weight: construct a weighted combination of donor countries tracking pre-crisis Greece, and read post-2008 divergence as the crisis effect. On the only substantively defensible pre-period, 2003–2008, only six countries have complete coverage and the fit is poor, with donor weights collapsing onto two countries. Restricting the match to 2007–2008 produces a near-exact pre-period fit on two observations against roughly twenty-five free donor weights, which is not evidence of fit. We report the diagnostics and not the divergence estimate, which has no defensible interpretation.

> Multi-domain breadth failed the incremental criterion: adding it to the reference model worsened Greece's residual from 6.93 to 10.39 and reversed its sign conditionally. The reversal is left uninterpreted.
>
> *Limits.* a result about the specification, not about power; the sign reversal is deliberately left uninterpreted.

The multi-domain breadth measure failed differently: adding it to the reference model worsened Greece's residual and reversed its sign under conditioning. We leave the reversal uninterpreted. Post-hoc explanation of sign reversals in failed specifications is the mechanism by which failed specifications return.

6.7 Robustness

Four checks bear on the results above and we summarise them together. The wild cluster bootstrap with the null imposed is the binding condition for two constructs that clear multiplicity correction, which is the reason it is applied rather than conventional cluster-robust inference. Leave-one-out refits establish that no supported result depends on a single country. The between/within decomposition establishes that the accumulated results are cross-sectional, and we report the imprecision of the within estimates rather than treating them as nulls. And the two specifications in Section 6.5 establish the boundary of what the absorption result can bear.

Two checks that a reader might expect are absent, and their absence is deliberate. We do not report a specification chosen for producing the most favourable Greek residual, because that choice is available in both directions and is uninformative. And we do not report the synthetic-control divergence, because a construction whose donor pool collapses onto two countries yields a quantity with no defensible interpretation, however precise it appears.

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7. Discussion

7.1 Is this a reporting artefact?

The most economical account of everything above is that Greek respondents answer subjective questions more negatively than others. Section 5.6 addresses this partly, but from within a single instrument. A better test compares Greek responses across *domains*: a general negative tendency should depress all of them roughly equally, whereas a pattern concentrated in financial items points toward circumstances.

> Greece is consistently among Europe's worst countries on life satisfaction and consistently worst on financial hardship and expectations. The greater extremity of the financial indicators suggests domain specificity, but does not rule out a broader negative reporting tendency.
>
> *Limits.* Greece is second-WORST on life satisfaction by 2024, not middling; a broader negative reporting tendency is NOT ruled out; Greek life satisfaction ROSE over the observed period, 6.2 to 6.9; only its rank worsened, because other countries improved faster; the series begins in 2013, at the crisis trough, with no pre-crisis baseline.

The pattern is domain-specific, but the difference is one of degree rather than kind, and an earlier version of this analysis described it too favourably. Greece is worst in Europe on the financial indicators and close to worst on general life satisfaction. Two quantities must also be held apart: Greek life satisfaction *rose* over the observed period, from 6.2 to 6.9, while Greece's *rank* worsened because other countries improved faster. A worsening rank on a rising series is not falling satisfaction.

DESCRIPTIVE CORROBORATION

FINANCIAL EXPECTATIONS AND LIFE SATISFACTION

The cross-domain comparison is descriptive corroboration for the domain-specificity reading rather than a test of it. Greece's financial indicators sit at the extreme of the European distribution while its general-wellbeing indicator sits close to it, and that difference in degree is what the comparison shows.

What may be concluded. The financial indicators are more extreme than the general-wellbeing one, which suggests domain specificity. A broader negative reporting tendency is NOT ruled out.

Limitation. Describing Greece as ordinary or middling on life satisfaction: it is second-worst in the EU by 2024. And reading a worsening RANK as falling satisfaction, when the Greek level rose over the period.

reporting_style_cross_indicator.csv, this project Greece ranks 1st of 27 on hardship and financial expectations, 2nd-6th on life satisfaction.

Two features of this test limit what it can deliver. A domain-specific pattern is consistent with genuine financial difficulty, and equally consistent with genuine financial difficulty accompanied by a general negative tendency; separating those would require either a pre-crisis baseline on the same instrument or an external anchor on Greek response style, and we have neither. And the comparison is between ranks in a European distribution, which move when other countries move: the Greek life-satisfaction rank worsened over a period in which the Greek level rose, because other countries improved faster. We therefore report levels alongside ranks throughout and treat the two as distinct quantities.

7.2 A pre-crisis baseline

The Eurostat wellbeing series begins in 2013, at the trough of the Greek crisis, and therefore cannot establish whether Greece was already unusual before 2008. We address this with a separate instrument, reported separately and never joined to the EU-SILC series.

Using publicly published weighted response distributions from the European Social Survey for six rounds in which Greece participated, and holding fixed the twelve countries present in all six, Greece sat approximately 0.8 points below the median *before* the crisis, fell to 5.64 by 2010/11, and recovered its level to approximately its pre-crisis value by 2023/24. Its comparative position did not recover: the gap to the median is wider than before the crisis, and Greece has ranked lowest of the twelve since 2010/11.

These means are reconstructed from displayed percentages and are approximate at that precision; no standard error can be attached to them and no test is run on them. The all-country ranks are not compared across rounds, since the ESS country set varies from twentytwo to thirty and a rank against a changing set would confound Greece's position with participation. The implication for Section 7.1 is direct: a low Greek baseline pre-dates the crisis, so the position cannot be attributed to the crisis alone, and a longstanding low-wellbeing pattern remains plausible without being established.

DESCRIPTIVE CORROBORATION

PRE-CRISIS WELLBEING BASELINE (ESS)

Across six rounds, holding the same twelve countries fixed, the Greek level fell and then recovered while the comparative position did not.

What may be concluded. On a balanced set of 12 countries observed in every Greek ESS round, Greece already sat about 0.8 points below the median before the crisis, fell to 5.64 by 2010/11, and recovered its LEVEL to roughly the pre-crisis value by 2023/24. Its comparative position did not recover: the gap to the balanced median is wider than before the crisis, and Greece has been worst of the 12 since 2010/11. A longstanding low-wellbeing pattern therefore remains plausible and generic pessimism or reporting culture CANNOT be dismissed.

Limitation. Splicing ESS to the Eurostat EU-SILC series, or reading the two as one trajectory. Attaching any confidence interval, standard error or significance test to these means, which are approximate reconstructions from displayed weighted percentages. Comparing ALL-COUNTRY ranks across rounds, since the country set varies from 22 to 30. Treating the decade after 2010/11 as observed.

European Social Survey Data Portal, public Analysis tab for stflife (life satisfaction, 0-10), rounds 1, 2, 4, 5, 10 and 11. Weighted distributions with the post-stratification weight; country means reconstructed by the authors. link

7.3 Factors this design cannot adjudicate

Several factors prominent in accounts of the Greek crisis are not established here. Some were examined: migration was tested on this panel and the cross-domain and ESS comparisons are descriptive analyses conducted for this paper. None of them can carry a headline finding, and we set out what each permits.

CONTEXTUAL EVIDENCE

INSTITUTIONAL TRUST

Institutional trust is low in Greece by OECD measurement, and a route by which it could matter is available: households not expecting effective support may experience identical material circumstances as more threatening. Trust is measured on a different instrument, sample and periodicity from the panel used here, and no defensible country-year merge exists for this period.

What may be concluded. May affect how households interpret insecurity; available evidence does not identify an independent effect.

Limitation. Presenting trust as an explanation of the residual, or implying it was tested and found to matter.

OECD (2024), OECD Survey on Drivers of Trust in Public Institutions - 2024 Results, Country Notes: Greece. OECD Publishing, Paris. Fieldwork October-November 2023, 30 OECD countries. 32% of Greek respondents report high or moderately high trust in central government against an OECD average of 39%. link

LITERATURE-GROUNDED CONTEXT

CRISIS AND ADJUSTMENT POLICIES

The adjustment programmes from 2010 reshaped incomes, employment protection, pensions and public services simultaneously and over a compressed period. They are the historical setting for every accumulated measure in Section 6.3.

What may be concluded. Help explain the historical setting; this project does not estimate their causal contribution.

Limitation. Attributing any share of the accumulated exposure to a specific programme or measure.

Andriopoulou, E., Kanavitsa, E. & Tsakloglou, P. (2020), Decomposing Poverty in Hard Times: Greece 2007-2016. LSE GreeSE Paper No. 149. EU-SILC microdata, ELSTAT waves 2008-2017. Anchored FGT0 on a 2007 base reaches 48% at the 2013 peak. link

CONTEXTUAL CONSEQUENCE

MIGRATION

Emigration of working-age Greeks during the crisis is well documented and interacts with the labour-market measures in both directions: plausibly a consequence of prolonged labour-market damage, and plausibly a contributor to the composition of who remains.

What may be concluded. A plausible consequence of prolonged labour-market damage and a possible contributor to it. It is UNSUPPORTED as an independent aggregate predictor here.

Limitation. Reading it as a driver, or as ruled out. E3 tested it on this panel and found nothing (p=0.4006), which speaks to aggregate prediction and not to either causal direction.

Lazaretou, S. (2016), The Greek brain drain: the new pattern of Greek emigration during the recent crisis. Economic Bulletin, Bank of Greece, issue 43, pp. 31-53. 427,000 residents aged 15-64 left permanently 2008-2013; ~223,000 of them aged 25-39. The E3 null (p=0.4006) is this project's own evidence. link

FUTURE HYPOTHESIS

TAX BURDEN AND UNEQUAL TREATMENT

Published incidence research establishes that the Greek indirect tax system became markedly more regressive across the crisis. If household disposable positions worsened through taxation in ways income-based poverty measures capture poorly, that would be one route to a gap of the kind documented here.

What may be concluded. Published incidence evidence establishes that Greece's indirect tax system became markedly more regressive over the crisis. Whether that channel contributes to the hardship gap is a plausible hypothesis and is NOT tested here.

Limitation. Any quantitative statement linking tax burden to the hardship gap. The literature is about incidence, not about this outcome, and this project tested nothing on tax.

Kaplanoglou, G. (2015), Who Pays Indirect Taxes in Greece? From EU Entry to the Fiscal Crisis. Public Finance Review 43(4), 529-556. Microsimulation on Household Expenditure Survey data, 1988-2011. The 2011 indirect tax system is the most regressive of the period, with the sharpest effects on families with children and the unemployed. link

7.4 Implications for measurement

AUTHOR INTERPRETATION

POLICY IMPLICATIONS

The analysis shows repeatedly that no single official indicator captures what Greek households report. AROP misses both concept and yardstick; AROPE adds breadth but a declining share of it; anchored poverty registers what the relative line cannot; and accumulated labour and housing exposure carries information none of the present-day measures hold.

What may be concluded. POLICY RECOMMENDATION, NOT AN EMPIRICAL CONCLUSION: poverty dashboards should combine AROP, its real threshold, anchored poverty, AROPE/deprivation and accumulated labour and housing indicators.

Limitation. Presenting this as a finding. It follows from what the analysis showed AROP alone misses; it is not itself a result.

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8. Limitations

These are properties of the design rather than qualifications appended to it, and several of them bound the results more tightly than the results themselves.

Aggregate inference. The unit of analysis is the country-year. Every result is an aggregate association, and an aggregate relationship need not hold at household level. That accumulated unemployment predicts national hardship rates does not establish that individuals who experienced unemployment report more hardship.

No causal identification. Nothing in this design identifies a causal effect. We report associations conditional on AROP and year effects, and the direction of any underlying mechanism is not established.

Cluster count. Twenty-seven clusters is marginal for inference, which is why the bootstrap condition exists and why so many constructs land as inconclusive. A larger panel would convert a substantial part of this paper from silence into evidence.

Short outcome window. The accumulated measures reach back to 2008 or 2010, but the outcome supports roughly a decade of variation, and within-country identification draws only on that. This is the direct cause of the imprecision in Section 6.4.

Shared instrument. The corroboration in Section 5.6 comes from items collected in the same interview as the outcome, and Section 6.5 shows that admitting such a predictor moves Greece's residual across the full range of the sample.

Unexplained remainder. Of the 52.6-point average distance, the accounts established here address a minority. We do not attribute the rest.

External validity. Greece is studied here because it is the extreme case, and that is also the principal constraint on generalisation. The inferential results are estimated across twenty-seven countries and are not Greece-specific — the leave-one-out refits confirm that dropping Greece does not overturn them — but the question the paper asks is prompted by a divergence that only Greece exhibits at this magnitude. Whether the same accounts would decompose a smaller divergence elsewhere is untested. We would expect the measurement components to travel, since they follow from how the indicators are constructed rather than from Greek particulars, and would expect the accumulated-exposure results to travel only to countries that experienced a contraction of comparable duration. Neither expectation is established here.

Two further limitations concern the outcome itself. The item asks about difficulty making ends meet without specifying a reference period or a standard, so the quantity respondents report is not fixed by the instrument; this is a general property of subjective items and is the reason Section 7.1 treats the reporting question directly rather than assuming it away. And the outcome is a household-level assessment aggregated to a national share, so a country with a small number of households in extreme difficulty and a country with widespread moderate difficulty can return the same figure. The measures tested here are not able to distinguish those cases, and no result in this paper should be read as bearing on the depth of hardship as opposed to its incidence.

9. Conclusion

Greek households report financial difficulty far above what the official relative-poverty rate predicts, persistently and by a margin that separates Greece from the European distribution rather than placing it at one end of it. Three bodies of evidence make part of that distance more intelligible.

The official measures are narrower than the experience they are taken to summarise. The broader AROPE measure closes about a fifth of the distance and its contribution is declining, and the relative income line fell with the Greek economy, which a relative measure cannot register by design. What households report corresponds to concrete affordability failure, within countries as well as across them, though this is corroboration from inside one survey instrument rather than independent validation. And both present conditions and accumulated exposure carry predictive information beyond income poverty: material resources, long-term unemployment and wage-adjusted affordability at the present-day stage, and accumulated unemployment, the duration of wage non-recovery and housing-cost deterioration conditional on their own present-day counterparts.

What the evidence does not support is equally part of the result. No causal claim is made anywhere. No within-country dynamic reading is supported, and the within-country tests are too imprecise to exclude one. Most constructs that did not clear the conditions are unresolved rather than excluded. A central absorption result reverses under a defensible alternative specification. A broader negative reporting tendency is not ruled out, and the ESS comparison indicates a Greek deficit that pre-dates the crisis. And most of the gap remains unexplained.

Four developments would change these conclusions. Corroboration from outside EU-SILC — administrative arrears, utility disconnection records, or an independent survey — would settle whether the co-movement in Section 5.6 reflects shared substance or shared instrument, which Section 6.5 shows this design cannot resolve. A longer or wider panel would convert most of the inconclusive results into findings or genuine nulls. Household-level analysis would permit the within-country tests this design cannot support. And respondent-level access to a pre-crisis wellbeing instrument would place Section 7.2 on a footing that admits inference.

The measurement implication follows from what the single indicators miss rather than from any test reported here, and we state it as an interpretation: a poverty dashboard combining the relative rate, its real threshold, anchored poverty, deprivation, and accumulated labour and housing exposure would represent a prolonged contraction better than any one of them alone.

10. Data and reproducibility

All quantitative inputs are published series. The outcome, income poverty, at-risk-of-poverty-or-social-exclusion, deprivation and housing series come from Eurostat's EU-SILC collection; long-term unemployment from the Labour Force Survey; prices and wages from the harmonised price index and national accounts; and consumption per head from the purchasing power parity programme. Institutional trust figures in Section 7.3 come from the OECD's 2024 survey on drivers of trust in public institutions and were read from the primary release. The comparison in Section 7.2 uses publicly published weighted response distributions from the European Social Survey portal; respondent-level ESS files are not used, and no result in this paper depends on them.

The analysis runs end to end from these published sources by code, without manual steps. Every figure in this paper carries the numbers behind it in an expandable table beneath the chart, so each chart can be checked against its own data without recourse to the source files. A technical report and a statistical appendix accompany this paper and carry the full evidence base, including the constructs and specifications not reported here.

Analysis and drafting were assisted by a large language model under author direction. All specifications, decision rules and conclusions were fixed by the authors, and every quantitative statement in this paper is generated from the source data rather than transcribed.

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